10^{th} Set of Problems

1. For a certain *RR manipulator*, such as the one illustrated in Figure 1(a), the equations of motion are given by

$$\begin{pmatrix} 4+c_2 & 1+c_2 \\ 1+c_2 & 1 \end{pmatrix} \begin{pmatrix} \ddot{\vartheta}_1 \\ \ddot{\vartheta}_2 \end{pmatrix} + \begin{pmatrix} -s_2\left(\dot{\vartheta}_2^2 + 2\,\dot{\vartheta}_1\,\dot{\vartheta}_2\right) \\ s_2\,\dot{\vartheta}_1^2 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}.$$

Assume that the second joint is locked at some value ϑ_2 using brakes.

(a) What is the minimum and maximum inertia perceived at the first joint, as we vary the second joint's angle, ϑ_2.

(b) Assuming that the first joint is controlled with a *PD controller*,

$$\tau_1 = -40\,\dot{\vartheta}_1 - 400\,(\vartheta_1 - \vartheta_{1_d})\ ,$$

what are the maximum and minimum values of the closed loop frequencies and damping ratios, and at what values of ϑ_2 do they occur.

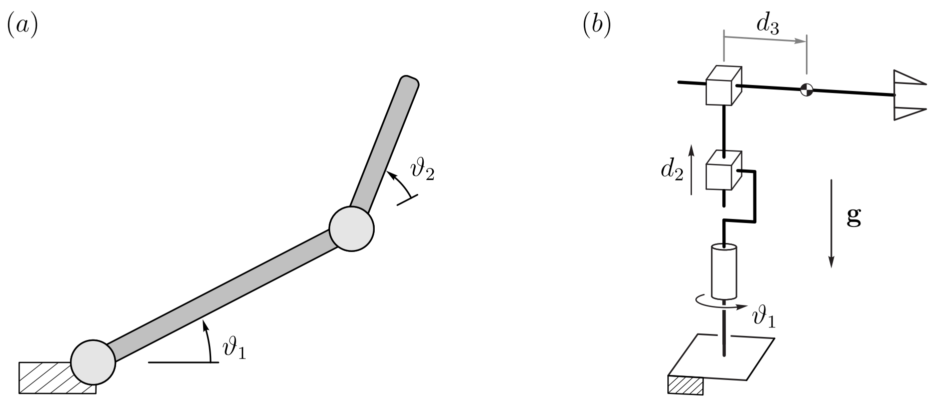

Figure 1: (a) Two-link planar arm. (b) Cylindrical manipulator.

2. Consider a cylindrical manipulator, such as the one illustrated in Figure 1(b), whose equations of motion have been written as

$$\begin{pmatrix} I_z^{(1)} + I_y^{(2)} + I_y^{(3)} + m_3\,d_3^2 & 0 & 0 \\ 0 & m_2+m_3 & 0 \\ 0 & 0 & m_3 \end{pmatrix} \begin{pmatrix} \ddot{\vartheta}_1 \\ \ddot{d}_2 \\ \ddot{d}_3 \end{pmatrix} + \begin{pmatrix} 2\,m_3\,d_3\,\dot{d}_3\,\dot{\vartheta}_1 \\ (m_2+m_3)\ |\mathrm{g}| \\ -m_3\,d_3\,\dot{\vartheta}_1^2 \end{pmatrix} = \boldsymbol{\tau}\,.$$

Design a controller for the manipulator, of the P-D type, with gravity compensation. Show the block diagram of the controlled system. Clearly indicate all assumptions made, and through the root locus comment the system performance in the presence of the nonlinearities.